

PVCELL–25 power cell

Rev C0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | 25 kW (27.5 kW max) bidirectional 4–switch buck–boost cell, 2 x 4 x SG2M040170HJ

01 CELL connector, logic supply, ID

40–way CELL contract to CTRL–C2000, +24V = safety–switched +24V_GD;
LMR36015 3.3 V + 5 V (driver logic side), TPS3710 on RDY, ID 4.99k, AGND tie

File: sheets/01_cell.kicad_sch

02 Fail–safe RS–422, fault latch

AM26LV32E receivers with external fail–safe bias (open = gate off),
FLT driver, OC/OT fault latch, EN_DRV, PWM gating

File: sheets/02_logic.kicad_sch

03 Power stage, decoupling, dampers

Leg A: S1/S2 on port A, leg B: S3/S4 on port B, 2 x SG2M040170HJ each;
film decoupling and RC damper per leg at the device pins

File: sheets/03_power.kicad_sch

04 Film banks, bleeders, inductor

Port A/B C4AQ banks + TNPV bleeders, bus studs to PV–PORT,
off–board CUSTOM inductor on studs SW_A / SW_B, winding NTC header

File: sheets/04_bus.kicad_sch

05 Inductor current, temperatures, trips

AN1 = LEM LA 150–P + driver, AN2/AN3 = heatsink/winding temperature,
OC window + heatsink OT comparators, 2.5 V reference, NTC pass–through

File: sheets/05_sense.kicad_sch

06 Gate drive S1 (leg A)

gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM2, DESAT to the drain

File: sheets/06_gd_s1.kicad_sch

07 Gate drive S2 (leg A)

gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM1, DESAT to the drain

File: sheets/07_gd_s2.kicad_sch

08 Gate drive S3 (leg B)

gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM4, DESAT to the drain

File: sheets/08_gd_s3.kicad_sch

09 Gate drive S4 (leg B)

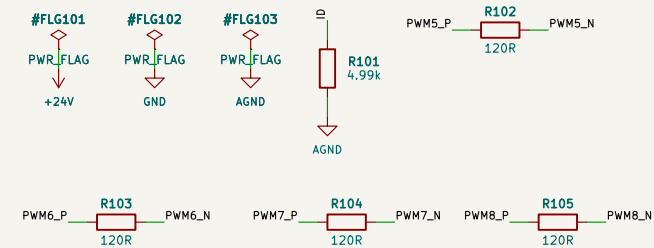
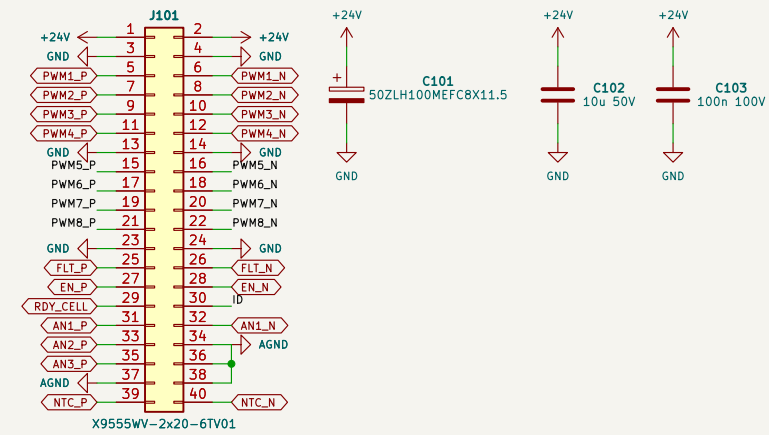
gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM3, DESAT to the drain

File: sheets/09_gd_s4.kicad_sch

CALCULATED, not measured: design_check() asserts every number on each build (outputs/PVCELL–25_design_check.txt); component values are read from sim/out/pv_design/cell_spec.json.
Local trips: inductor OC +/-72.2 A (68.5–76.0 A), gates off 0.94 us (PWM) / 1.73 us (EN) vs 3.66 us allowed; heatsink OT 95.0 C (92.3–97.9 C), open probe trips; DESAT per driver; CTRL backup peak 106.0 A.
AN1 = LEM LA 150–P on inductor lead A: 20.0 mV/A differential, CM 1.25 V, < 1 ohm, linear +/-114 A, 0.52 us, 0.0759 A/LSB at the CTRL. Bus OV: no local comparator (CTRL path 26 us -> 1115 V, 0.835 x V_DSS).
Fail–safe off: open, unplugged, unpowered or tri–stated pairs read <= –0.29 V = LOW; RDY push–pull; +24V removal drops RDY at >= 19.7 V; +3V3 or +5V loss -> drivers off (VCC1 UVLO / EN low). No bootstrap: 100 % duty OK.
Domains: PELV, HV, GD_S1..S4, PE. Isolators: NSI6651ASC–Q1 + UCC14241–Q1 x 4, LA 150–P, L_CELL. CONFORMAL COATING to PD1 (IEC 60664–3) over every HV–PELV barrier part and the power stage (IC–17, IC–22). 1000 V DC (ECO–10).
LAYOUT: loop <= 20 nH, bank <= 15 nH from each leg, damper loop <= 3 nH, clamp branch <= 1.5 nH, Miller clamp loop <= 1 nH, drain–pin slot. BENCH: Miller margin 0.53 V, LA 150–P dv/dt, damper power, pad PD test.

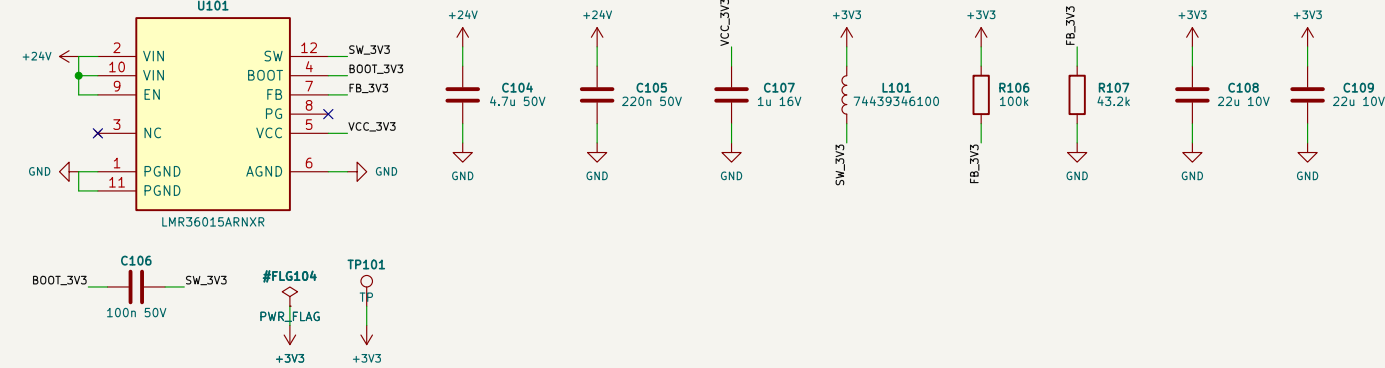
Not bench–validated. Values: sim/out/pv_design/cell_spec.json; checks: design_check() Sheet 1 Schematics authored by Chinmoy Bhuyan HV–PELV reinforced 1000 V DC (ECO–10); barriers listed on the cover		
Sheet: / File: PVCELL–25.kicad_sch		
Title: PVCELL–25 power cell – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: C0
KiCad E.D.A. 10.0.6	Id: 1/10	

CELL CONNECTOR (INTERFACES.CELL)



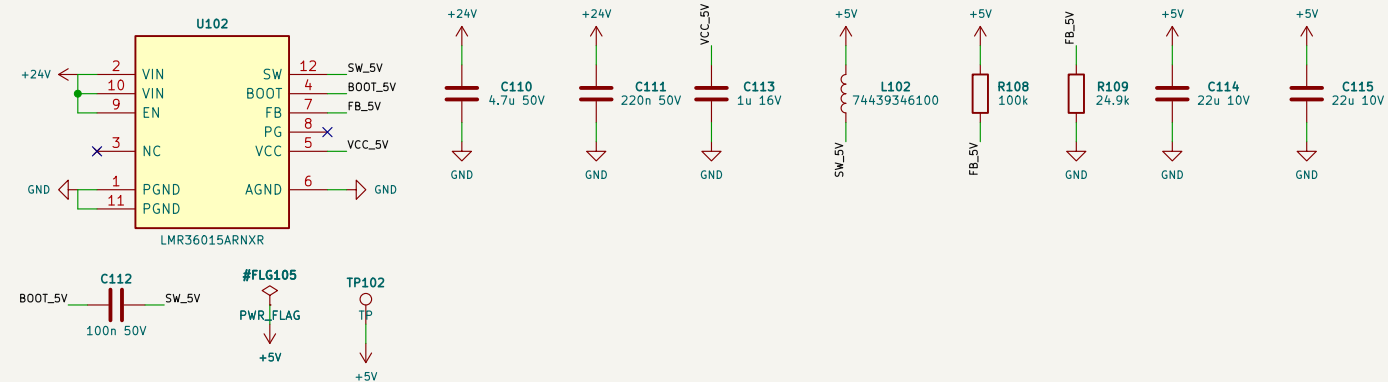
Ribbon 1:1 to a CTRL-C2000 cell port. AN2_N/AN3_N = AGND here (single-ended 0-2.5 V, AN_N tied at the source per the CTRL AFE). NTC_P/N float to the second heatsink NTC (biased on CTRL). PWM5-8 unused: 120R only. +24V: 0.52 A of 1.0 A worst case (design_check); <= 600 uF on the rail.

LOGIC SUPPLY +3V3 (LMR36015, 400 KHZ)



24 V -> 3.315 V (100k/43.2k, SNVSB49D 10.2, as PV-PORT). Runs down to 4.2 V input, so the logic keeps every gate disabled while +24V collapses. Load <= 119 mA (datasheet maxima).

LOGIC SUPPLY +5V (LMR36015, 400 KHZ)



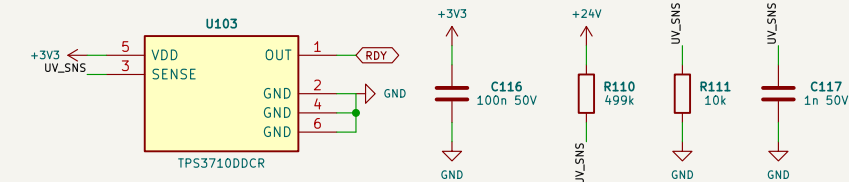
24 V -> 5.02 V (100k/24.9k): VCC1 of the four NSI6651 and the AHCT input stage - the driver input thresholds are specified at VCC1 = 5 V only (p7). Load <= 39 mA (datasheet maxima).

AGND – GND STAR POINT



Analog return (LA 150-P M / RM, +/-15 V COM, op amps, comparators, reference, NTCs, ID) meets GND only here.

+24V SUPERVISOR (TPS3710) ON RDY



OUT is open drain on the RDY line: RDY low below 19.66–20.40 V, released above 20.12–20.60 V. That is above the UCC14241–Q1 UVLO (17.1–18.9 V falling): EN_DRV drops and every driver pulls its gates to VEE while the bias is still regulating → clean turn-off when the safety chain removes +24V_GD (D-012).

Not bench-validated. Values: sim/out/pv_design/cell_spec.json; checks: design_check()
Sheet 2
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced 1000 V DC (ECO-10); barriers listed on the cover

Sheet: /01 CELL connector, logic supply, ID/
File: 01_cell.kicad_sch

Title: PVCELL-25 power cell – CELL connector, logic supply, ID

Size: A3

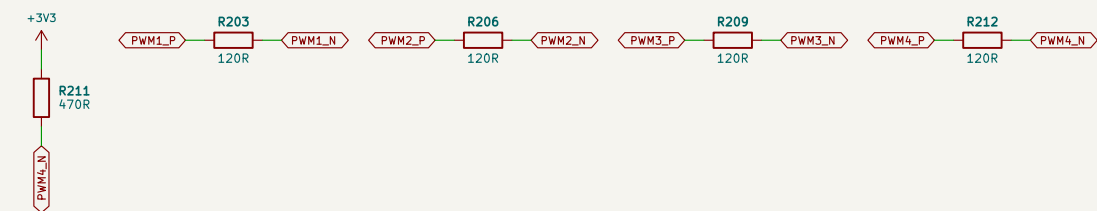
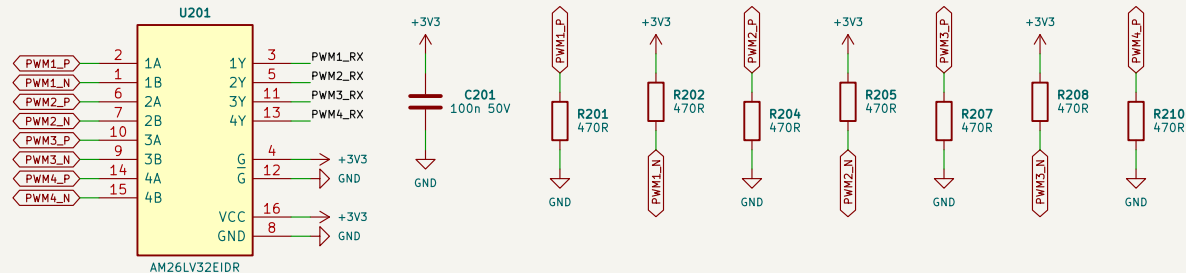
Date: 2026-10-04

Rev: C0

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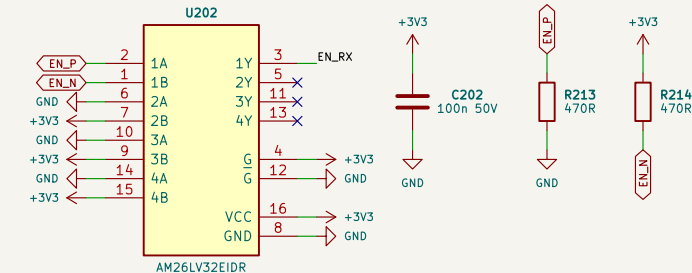
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PWM1-4 RECEIVERS (FAIL-SAFE)



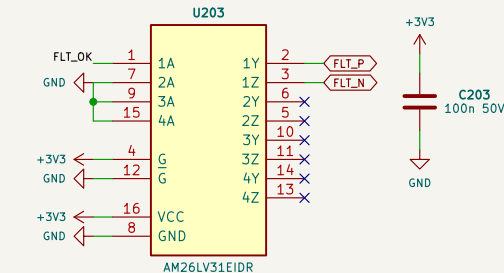
A = PWMk_P, B = PWMk_N. 470R P->GND, 470R N->3V3, 120R across: an open, unplugged or unpowered pair reads -0.30 V / -0.29 V worst case (< VIT- -0.2 V) -> output LOW = gate off. The receiver's own open-input fail-safe (output HIGH) never applies: the bias always defines the pair. Driven high: 1.9 V.

EN RECEIVER (FAIL-SAFE)



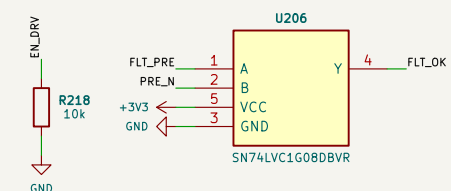
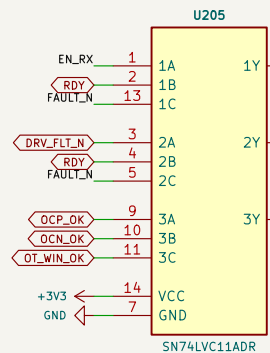
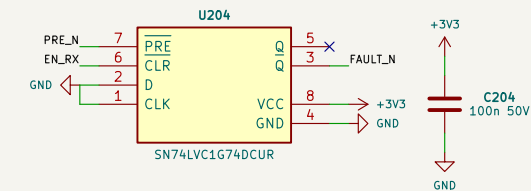
Same bias: open EN = LOW = drivers disabled. Spare receivers: A low, B high.

FLT DRIVER



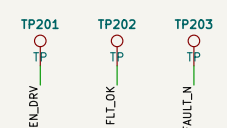
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FLT_P = FLT_OK, FLT_N = /FLT_OK, CTRL receives the pair swapped with its own bias: healthy = -2 V
= no fault; this board unpowered = AM26LV31E outputs HI-Z (SLLS848C) -> CTRL bias -> fault.
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FAULT LATCH, GATE ENABLE, FLT, RDY



PRE_LN = QCP_OUT and QCN_OUT AND OT_WIN_OK (low on fault) sets the latch: FAULT_LN; low until EN drives; with EN low and a fault still present both /PRE and /CLR are $Q = Q/H$, (Table 8-1), so FLT_OK = DRV_FLT_LN and RDY and FAULT_LN and PRE reports the LIVE fault (PWR-OT (INT-17): EN rising re-sets the latch. EN_DRV = EN_RX and RDY and FAULT_LN drives every RST/EN_RDY (wired-AND; drivers, bias PG, +24V and +15V monitors) reaches the CELL pin through a push-pull SN74LVC1617: actively high when ready, low otherwise: unpowered = loff, the card's 100 k pull-down reads not ready (INT-14).

FAIL-SAFE OFF: THE THREE LOSS CASES

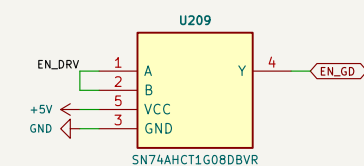
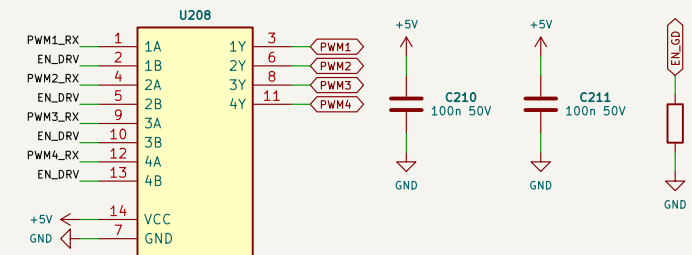


1 CABLE UNPLUGGED: no +24V_{cc} no gate bias, no logic: every gate sits at 0 V on its 10k gate-source resistor (VGS(th) ≈ -1.9 V). With +24V still present, an open pair reads -0.30 V → PWM and EN LOW.

2 CTRL UNPOWERED, or rev C card not enabling (if tri-state its PWM/EN drivers unless CHB_0K. Its enable and our RDY are high, INT-15: driver outputs Hi-Z (-100 uA, SLSB846 p5) → -0.29 V → LOW. In reset the card drives 0 (10k on its ePWM balls. EN = GATE+EN and CHB_0K and MCU enable).

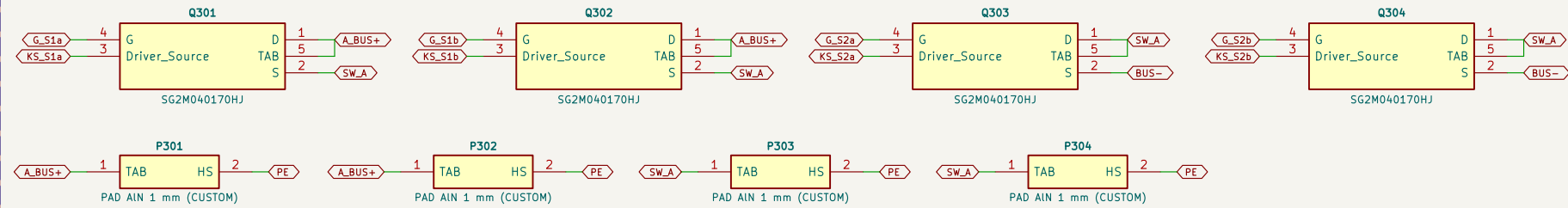
3 +3VS3 ABSENT: LVC/AM26 low → 10k pull-downs hold EN_DRV low → AHCT: EN_GD. PWMk low: RDY_CELL Hi-Z → card drives 0 V. FLT_HI-Z not pull-up: +5V ABSENT: N316651 VCC1 LVLO → OUT low (p28). UCC1D42-041 ENA = 0 → bias off. RDY / DRV_FLT_IN lose their pull-ups (to +5V) → not ready / recall. +24V RECOVERED (U0-122 channel B): RDY low at

PWM GATING + 5 V INPUT LEVEL (FAST PATH)



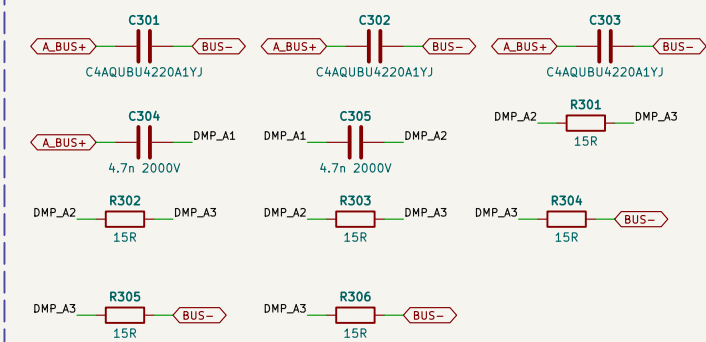
PWMk = PWMk_RX AND EN_DRV, EN_GD = EN_DRV, both at +5V: the NSI6651 input thresholds are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7), so VCC1 = +5V and the AHCT gates (VIH 2.0 V) lift the 3.3 V logic. A trip removes the commands within -20 ns, ahead of the RST/EN filter (0.48–0.8 us). +5V or +3V3 absent: EN_DRV / PWMk pulled down. AHCT unpowered -> 0. NSI6651 VCC1 UVLO -> OUT low (functional modes p28).

LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



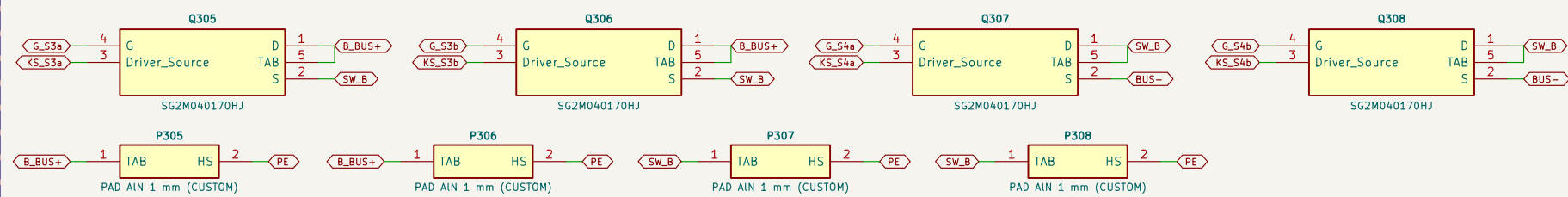
Kelvin source (pin 3) of each device returns to its own driver COM (gate-drive sheet).
Each tab on its own AIN ceramic (aluminum nitride) pad (P...: basic insulation HV-PE, IC-07; spec in the BOM). LAYOUT (IC-22):
milled slot between drain pin 1 and pins 2-4. SYMMETRY: equal drain/source busbar paths per switch.

LEG A DECOUPLING + RC DAMPER



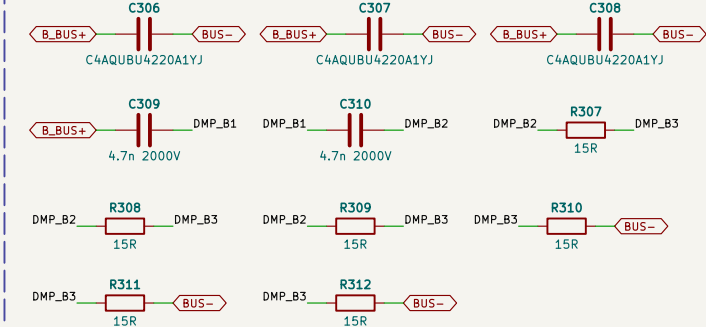
3 x C4AQUBU4220A1YJ at the S1/S2 pins: commutation loop cap-top-bottom ≤ 20 nH, bulk bank ≤ 15 nH from the leg (LAYOUT REQUIREMENTS, leg decd). Damper (PVR-02): 3.75 W normal / 5.68 W trip corner per leg into 2 x 3 CRCW2512-HP (7.4 W rated at 85 C, 500 V); damper loop at the device pins ≤ 3 nH.

LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



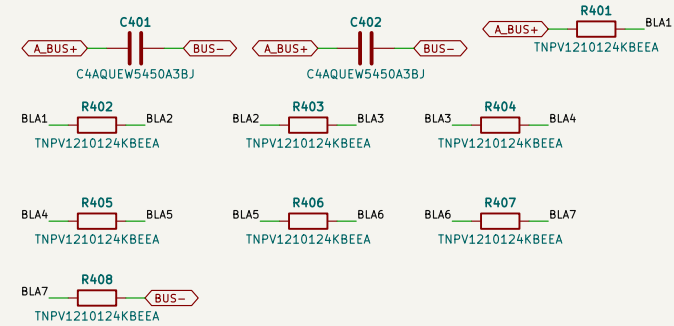
Kelvin source (pin 3) of each device returns to its own driver COM (gate-drive sheet).
Each tab on its own AlN ceramic (aluminium nitride) pad (P...: basic insulation HV-PE, IC-07; spec in the BOM). LAYOUT (IC-22):
milled slot between drain pin 1 and pins 2-4. SYMMETRY: equal drain/source busbar paths per switch.

LEG B DECOUPLING + RC DAMPER



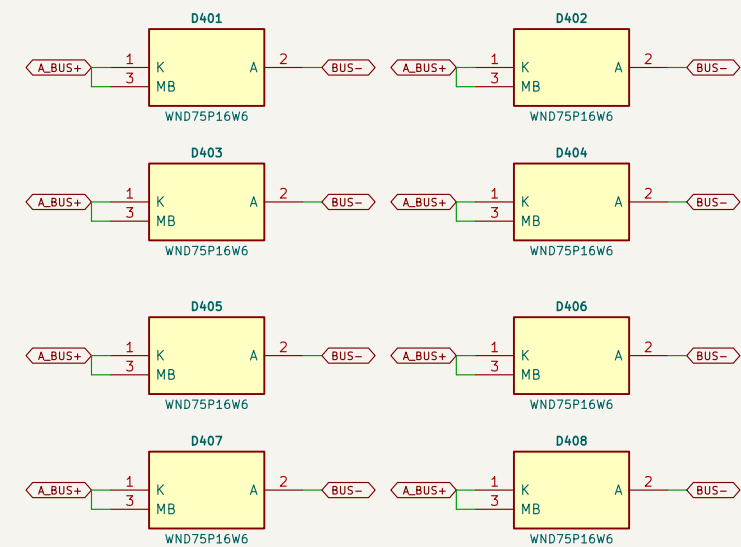
3 x C4Q4UBU4220A1YJ at the S3/S4 pins: commutation loop cap-top-bottom ≤ 20 nH, bulk bank ≤ 15 nH from the leg (LAYOUT REQUIREMENTS, leg deck). Damper (PVR-02): 3.75 W normal / 5.68 W trip corner per leg into 2 x 3 CRCW2512-HP (7.4 W rated at 85 C, 500 V): damper loop at the device pins ≤ 3 nH.

PORT A FILM BANK + BLEEDER



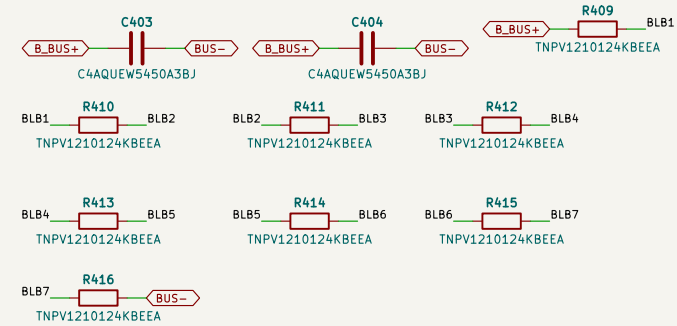
2 x C4AQEUW5450A3BJ = 90 uF, ripple 23.6 of 47.6 A rms (use limit). Bleeder 8 x TNPV1210 124k: 138 V and 152 mW per element at 1100 V; bank to 60 V in <= 5 min. Busbar to the leg <= 15 nH.

PORT A BANK REVERSE CLAMP (8 X WND75P16W6)



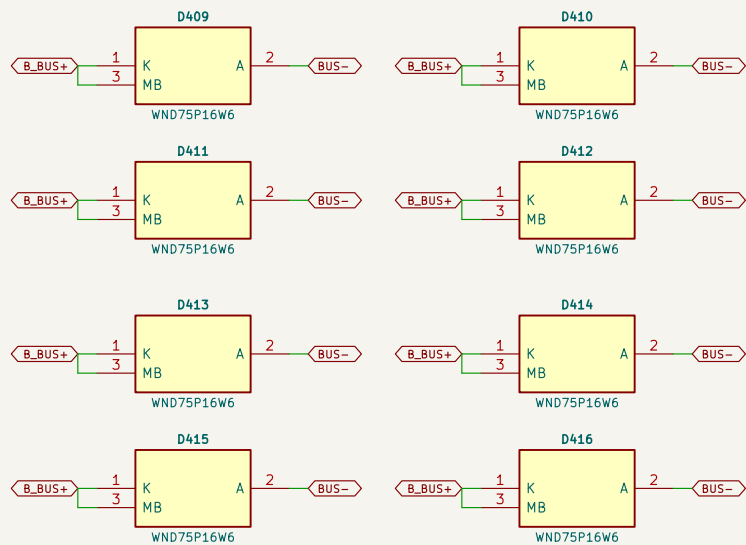
A bolted terminal short rings the bank below zero; these carry the freewheel instead of the SiC body diodes. Cathode to A-BUS+, anode to BUS-, in parallel. LAYOUT REQUIREMENT: on the laminated bus AT the C4AQ terminals, spread symmetrically between the capacitors, bank $\rightarrow$ group $\rightarrow$ bank ≤ 1.5 nH.

PORT B FILM BANK + BLEEDER



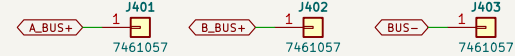
2 x C4QAUEW5450A3BJ = 90 uF, ripple 23.6 of 47.6 A rms (use limit). Bleeder 8 x TNPV1210 124k: 138 V and 152 mW per element at 1100 V; bank to 60 V in <= 5 min. Busbar to the leg <= 15 nH.

PORT B BANK REVERSE CLAMP (8 X WND75P16W6)



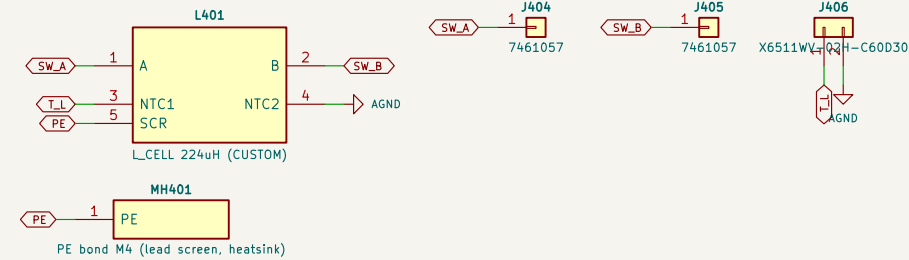
A bolted terminal short rings the bank below zero; these carry the freewheel instead of the SiC body diodes. Cathode to B_BUS+, anode to BUS-, in parallel. LAYOUT REQUIREMENT: on the laminated bus AT the C4AQ terminals, spread symmetrically between the capacitors, bank $\rightarrow$ group $\rightarrow$ bank ≤ 1.5 nH.

BUS TERMINALS TO PV-PORT



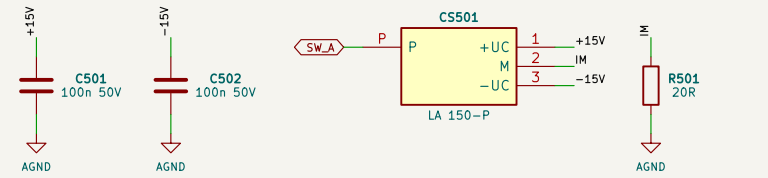
REDCUBE M3 studs, 100 A each: A_BUS+, B_BUS+, BUS- (net names as gen/port.py).

INDUCTOR (OFF-BOARD CUSTOM) + TERMINALS



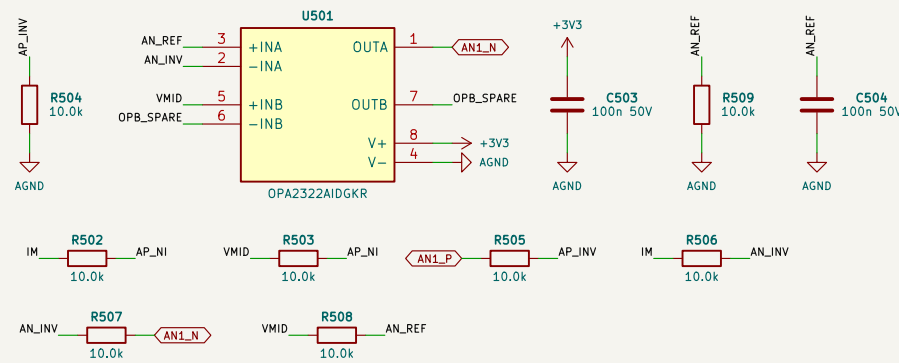
226 uH at 45 A (L0 244 uH), 37 turns on NPC290026 x 2; full spec in the part description / BOM.
Studs SW_A (lead A through the LA 150-P) and SW_B. Winding NTC (reinforced to the winding) on J.

AN1: INDUCTOR CURRENT (LEM LA 150–P AROUND INDUCTOR LEAD A)



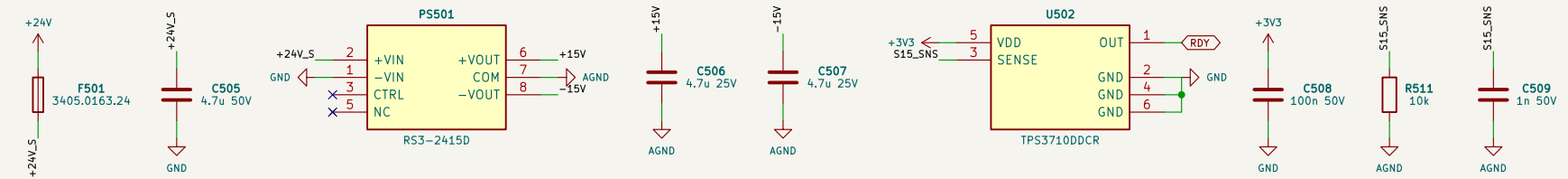
Lead A of L_CELL (Insulated, screen earthed in the aperture) passes through the transducer; positive = SW_A -> inductor -> SW_B. IS = IP/2000 into RM 20R to AGND; 10 mV/A. Linear +/-11A A delay <= 0.52 us; CTRL backup peak 106.0 A (< 129 A). LAYOUT: lead <= 100 C at the transducer (LA 150–P p4). RM and the +/-15 V return at one AGND point next to the transducer; no other conductor through the aperture.

AN1 DIFFERENTIAL DRIVER (20 MV/A, CM 1.25 V)



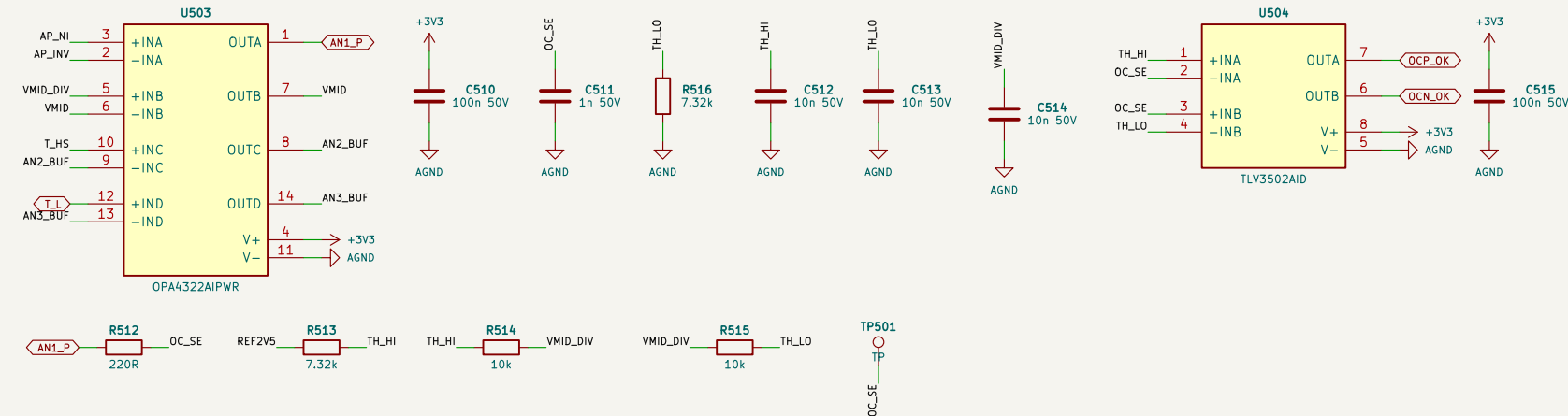
OPA4322 A: AN1_P = VMID + V_M (gain 2 on (V_M + VMID)/2); OPA2322 A: AN1_N = VMID - V_M (inverting, + at VMID/2). Direct drive (< 1 ohm closed loop) of the ribbon and the CTRL 1.50 k open-input bias (<= 2.3 mA). CTRL WFE 0.402 -> 0.0759 A/LSB. An unplugged cable reads far negative on CTRL (its bias).

+/-15 V FOR THE LA 150–P (RS3-2415D)



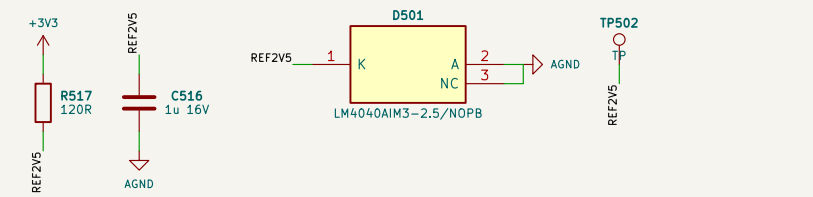
From +24V through a 0.5 A quick fuse (RECOM requires one); isolation of the module not used; COM = AGND. TPS3710 holds RDY low below 12.5–12.8 V on +15 V; a dead sensor supply never reads as 0 A while enabled. -15 V is not monitored separately (one converter); firmware checks AN1 plausibility.

OC WINDOW COMPARATOR +/-72.4 A (BOTH POLARITIES)



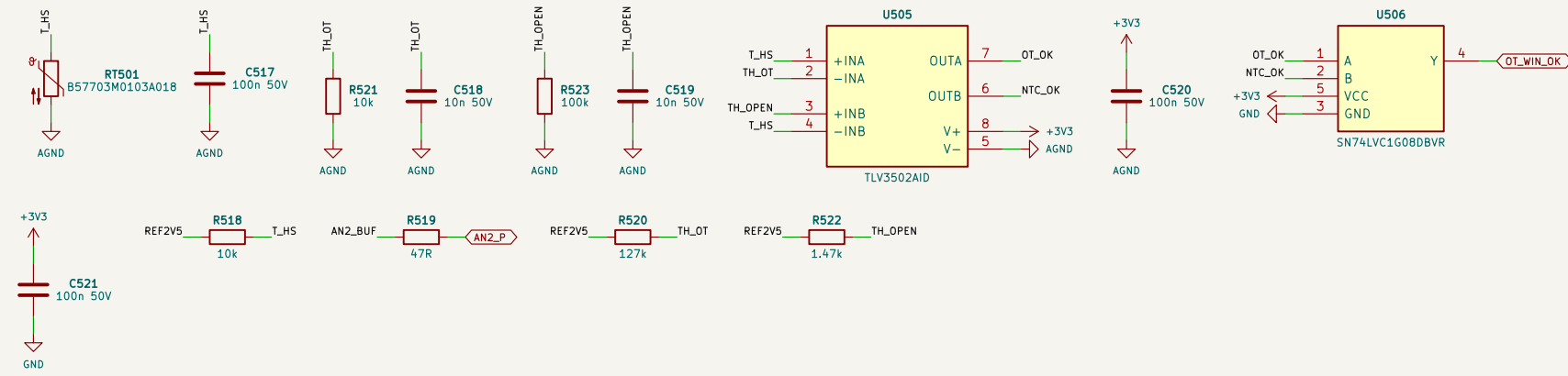
TLV3502 on AN1_P through 220R/1n; OCP_OK low above TH_HI, OCN_OK low below TH_LO -> latch. Trip 68.5–76.0 A (all tolerances); gate off 0.94 us after the crossing (PWM path), 1.73 us via EN; allowed 3.66 us. OPA4322 B: VMID buffer for the AN1 driver.

2.5 V REFERENCE



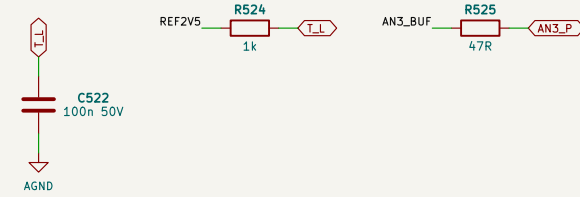
LM4040 A grade from 3V3 through 120R; 2.6 mA worst load + 65 uA, <= 8.4 mA total (15 mA max). Feeds the OC thresholds, the OT threshold and both NTC dividers (OT is reliometric).

AN2 HEATSINK TEMPERATURE + OT TRIP 95 C



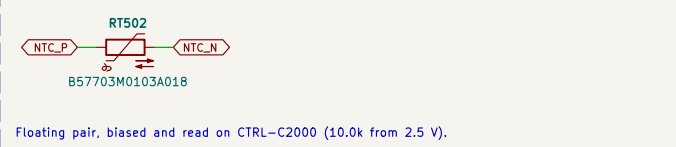
2.5 V - 10k - T_HS - NTC (B57703M, screwed next to the hottest device) - AGND; 2.43 V at -40 C ... 0.045 V at 150 C; buffered (OPA4322 C) to AN2_P via 47R. TLV3502 A: low above 95.0 C (92.3–97.9 C); B: low when T_HS > TH_OPEN (open probe reads 2.5 V; -40 C reads 2.432 V) -> OT_WIN_OK = OT_OK AND NTC_OK -> latch (PVR–08).

AN3 INDUCTOR WINDING TEMPERATURE



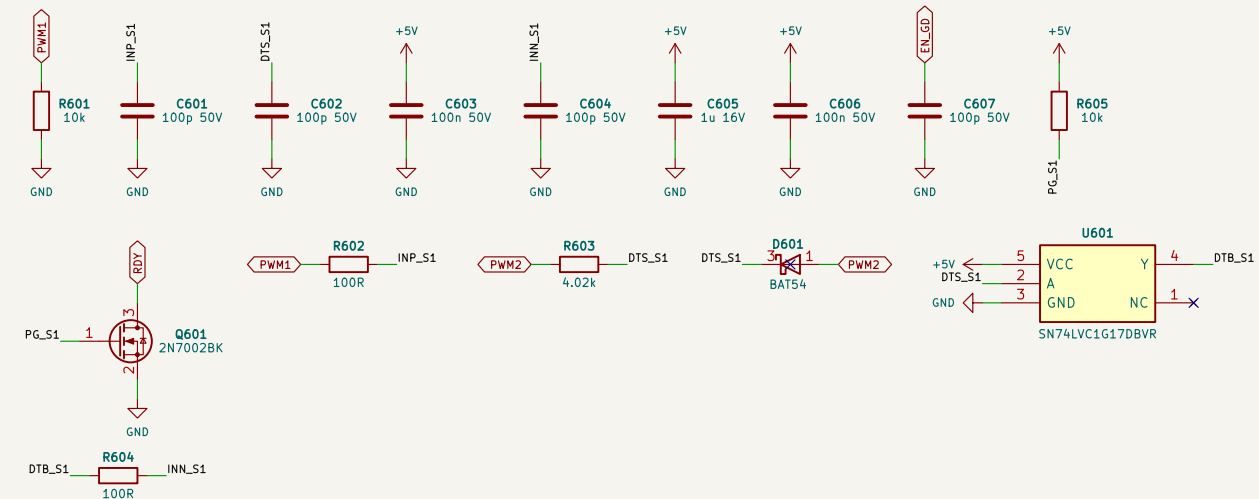
2.5 V - 1k - T_L - winding NTC (inside L_CELL header on sheet 04) - AGND; the NTC lead next to the 124 V/ns winding ends on AGND and the 100 nF T_L node; never on the reference. Buffered (OPA4322 D) to AN3_P via 47R; monitoring only (no local trip in the spec). Open NTC reads 2.5 V like -40 C; firmware plausibility check (winding colder than the heatsink at load = fault). Fan law and hot-spot limit in firmware.

NTC PAIR: SECOND HEATSINK NTC



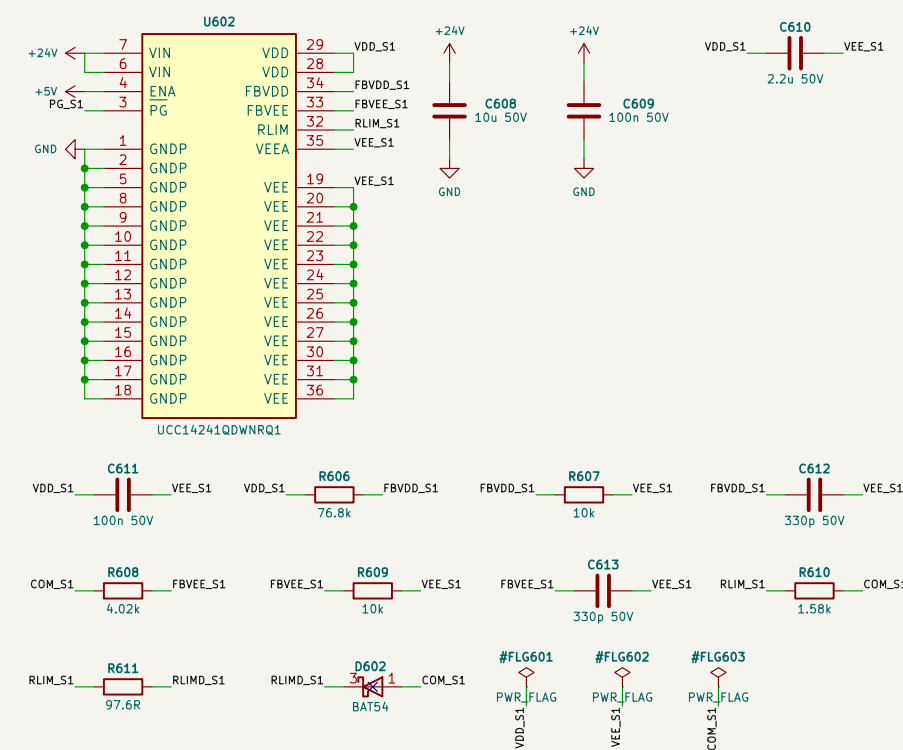
Floating pair, biased and read on CTRL–C2000 (10.0k from 2.5 V).

S1: LOGIC SIDE



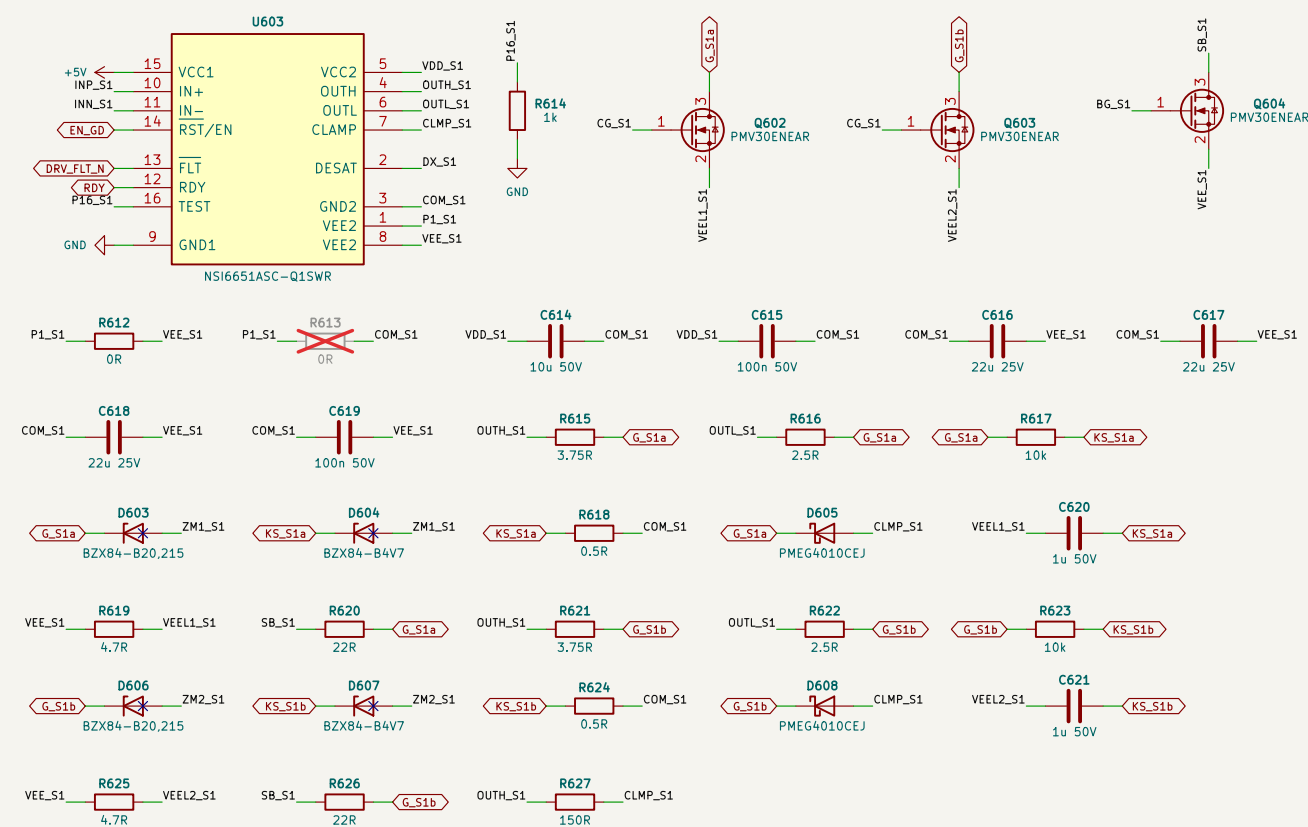
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

S1: ISOLATED BIAS +18 V / -3.5 V



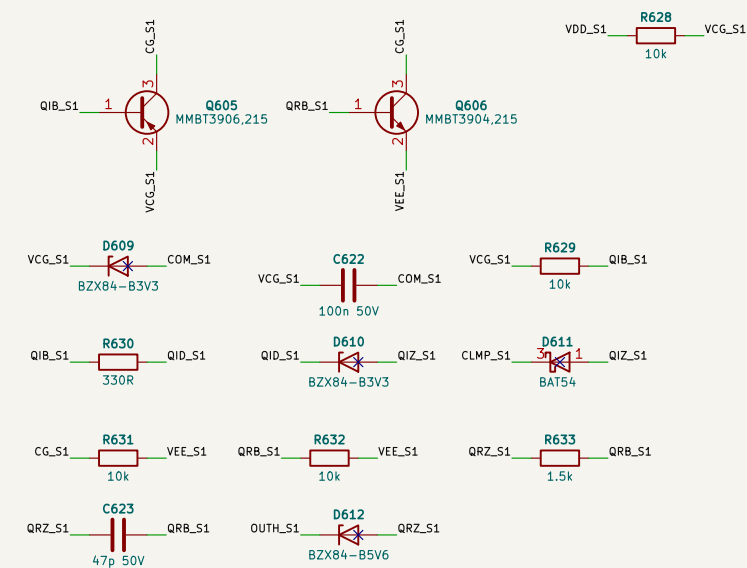
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

S1: DRIVER AND 2 GATES



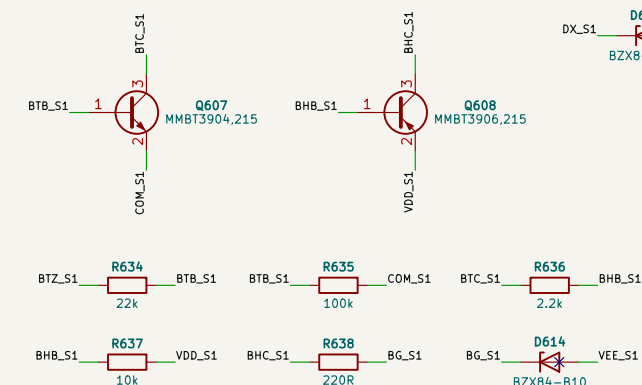
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV300NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

S1: MILLER CLAMP CONTROL



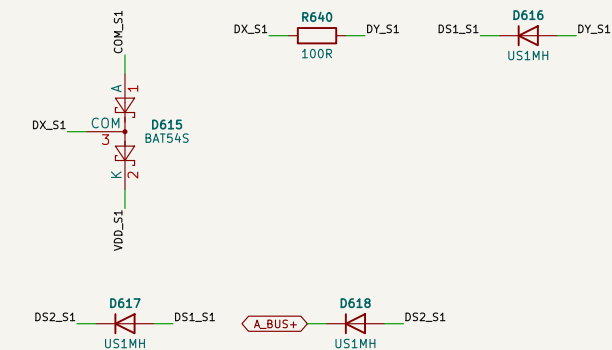
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

S1: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

S1: DESAT

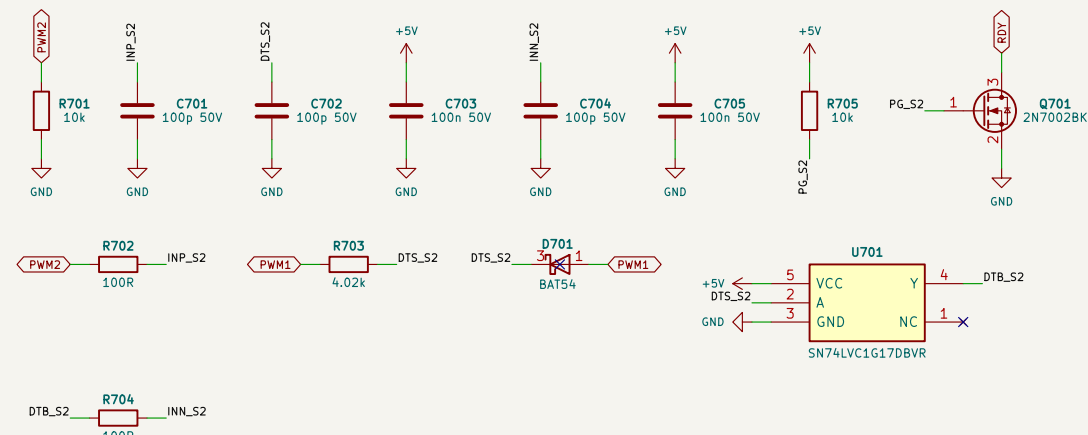


DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

S1: LAYOUT RULES + MILLER MARGIN

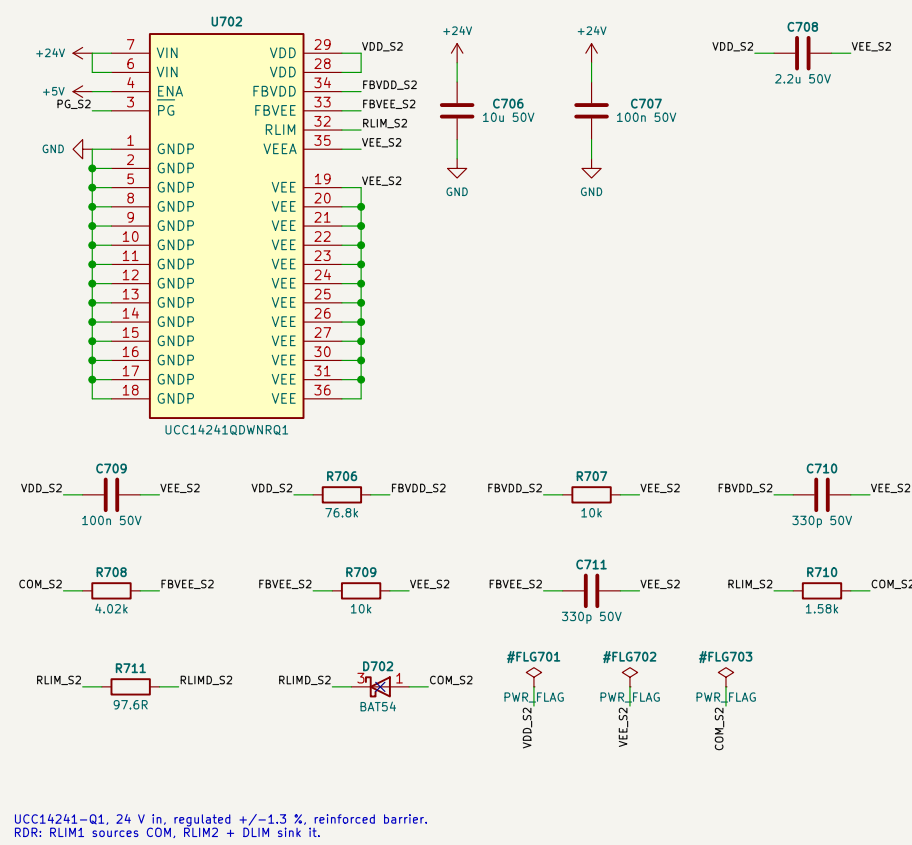
SYNTEK: Kevin and Kelvin: Si₂/Si₃ as coupled pairs of equal length (+/-10 %) from the COM_S1 star; Ron, Roff, Kelvin R: 10k; Zeners at each device; power sources joined only by the busbar. CLAMP LOOP: each AO3400A + C_VL at its device's gate-Kelvin pins, loop <= 1 nH, IC-22: milled spot between the drain pin and pins 2-4 of each TO-247-4; DESAT string <= 2.5 nH pad gap per diode; board coated to PD1. Miller hold (gdrv rev 5, ngpsic 1100 V / 72.4 A, 115 V_{gs}) die +1.4 V vs VGS(th) min 1.94 V at 175 °C, margin 0.5 V (drv rev 4 PNP: +2.14 V) – BENCH TEST (double pulse, hold, minimum-threshold devices).

S2: LOGIC SIDE



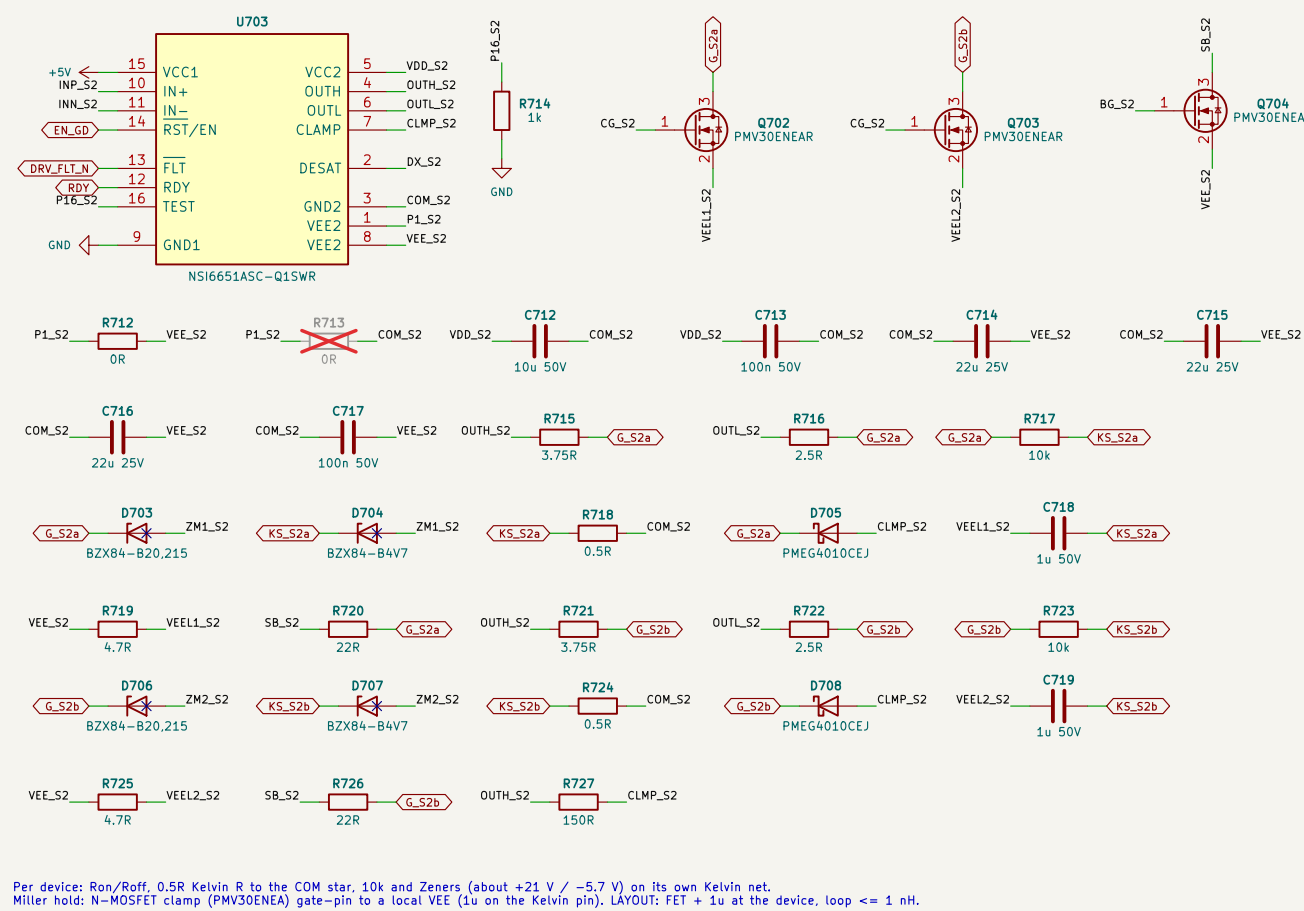
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

S2: ISOLATED BIAS +18 V / -3.5 V



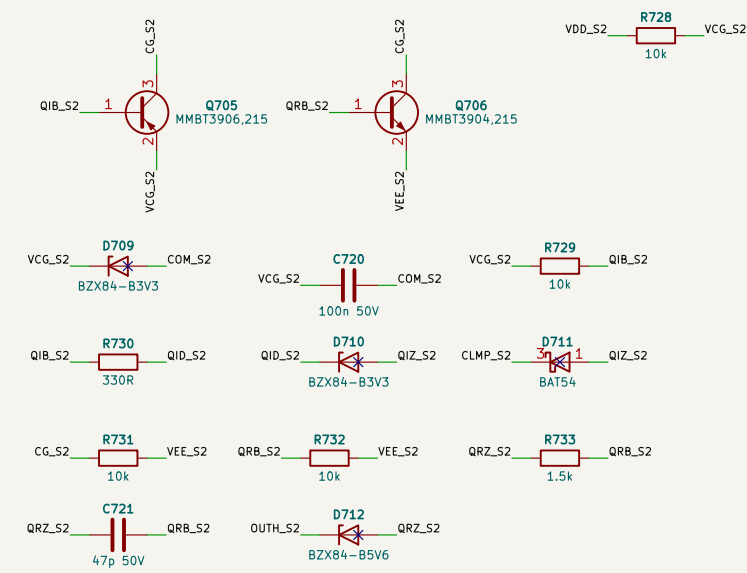
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

S2: DRIVER AND 2 GATES



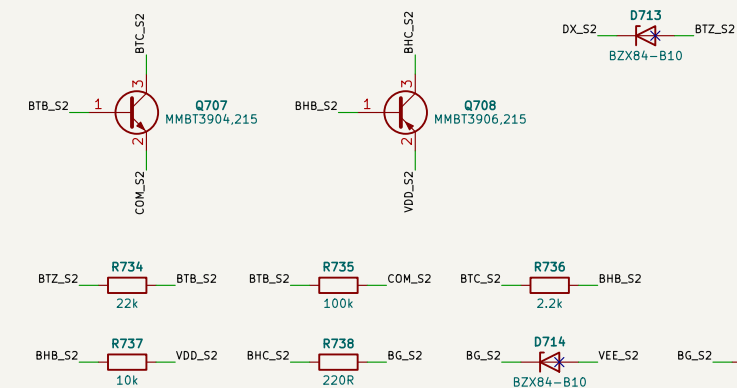
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

S2: MILLER CLAMP CONTROL



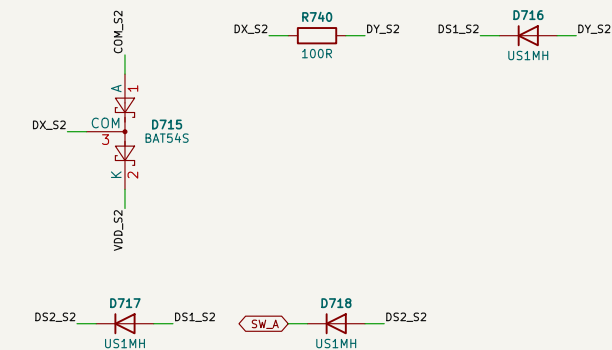
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

S2: DESAT

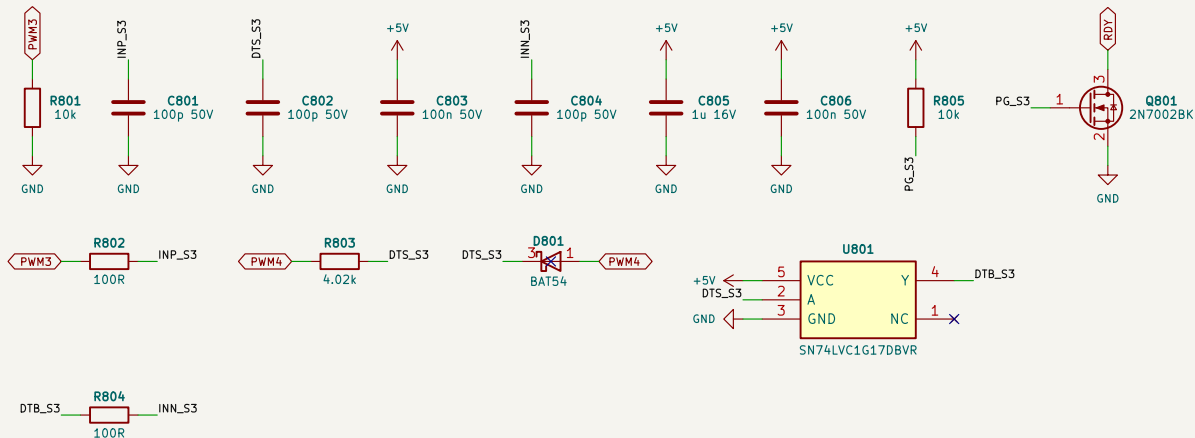


DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

S2: LAYOUT RULES + MILLER MARGIN

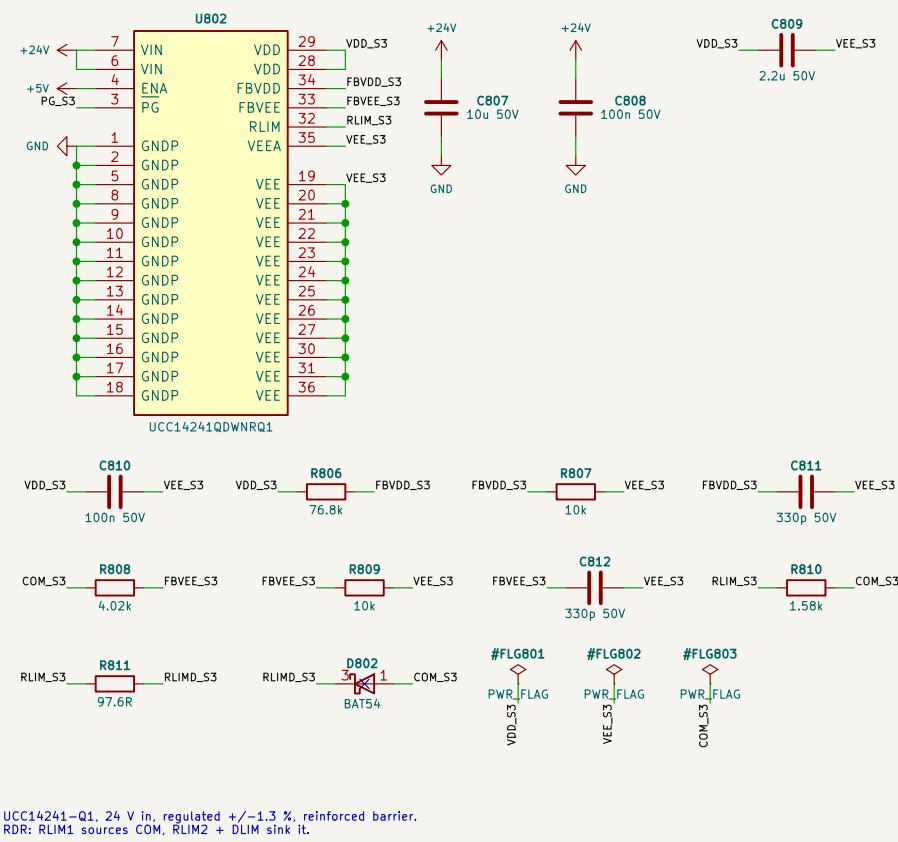
SYNTEK: Kevin and Kelvin: 52a/52b as coupled pairs of equal length (+/-10 %) from the COM_52 star; Ron, Roff, Kelvin R, 10k, Zeners at each device; power sources joined only by the busbar. CLAMP LOOP: each A03400A + C_VL at its device's gate-Kelvin pins, loop <= 1 nH, IC-22: milled slot between the drain pin and pins 2-4 of each T0-247-4L; DESAT string <= 2.5 m, pad gap per diode: board coated to PD1. Miller hold (gdrv rev 5, ngpsice 1100 V / 72.4 A, 115 V/ns); die +1.41 V vs VGS(th) min 1.94 V at 175 °C, margin 0.53 V (drv 4 PNP; +2.14 V) - BENCH TEST (double pulse, hold, minimum-threshold devices).

S3: LOGIC SIDE



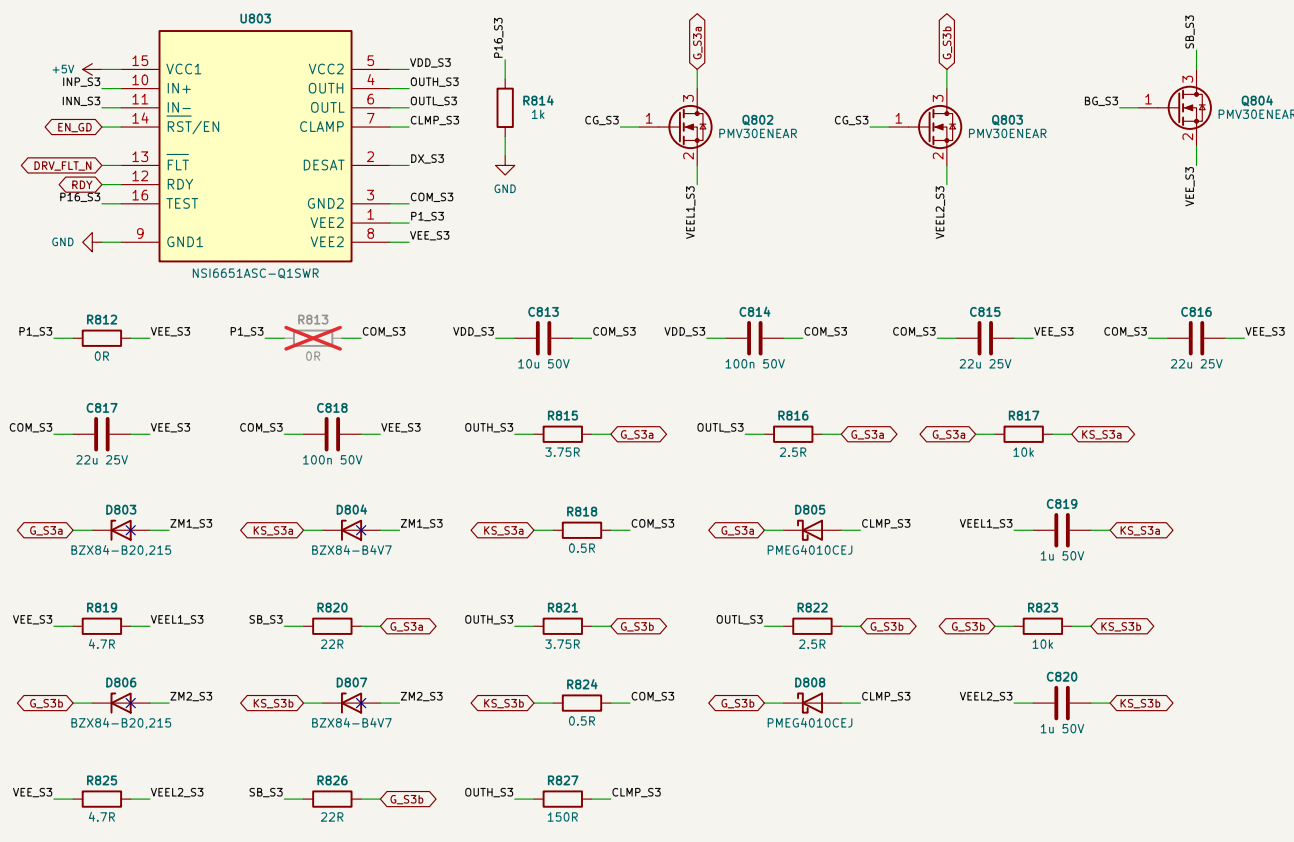
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock = min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

S3: ISOLATED BIAS +18 V / -3.5 V

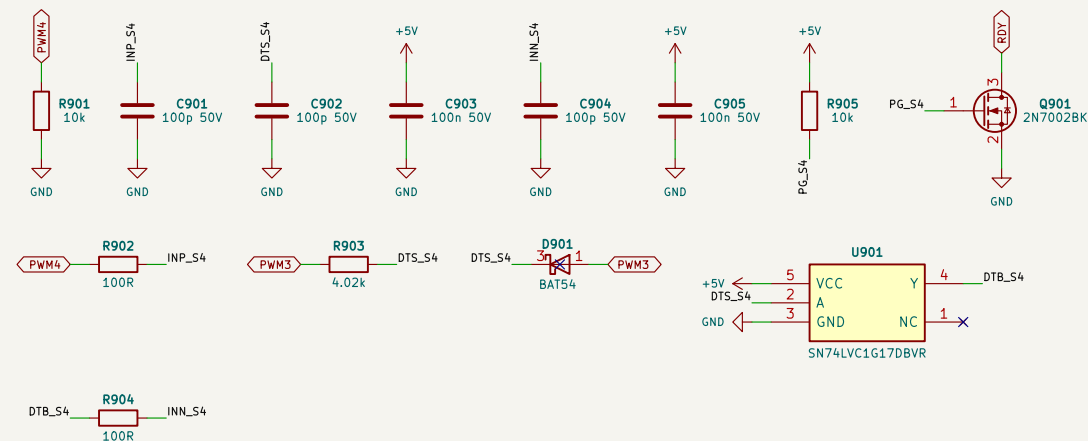


UCC14241-Q1, 24 V in, regulated +/-1.3 %, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLM sink IL.

S3: DRIVER AND 2 GATES

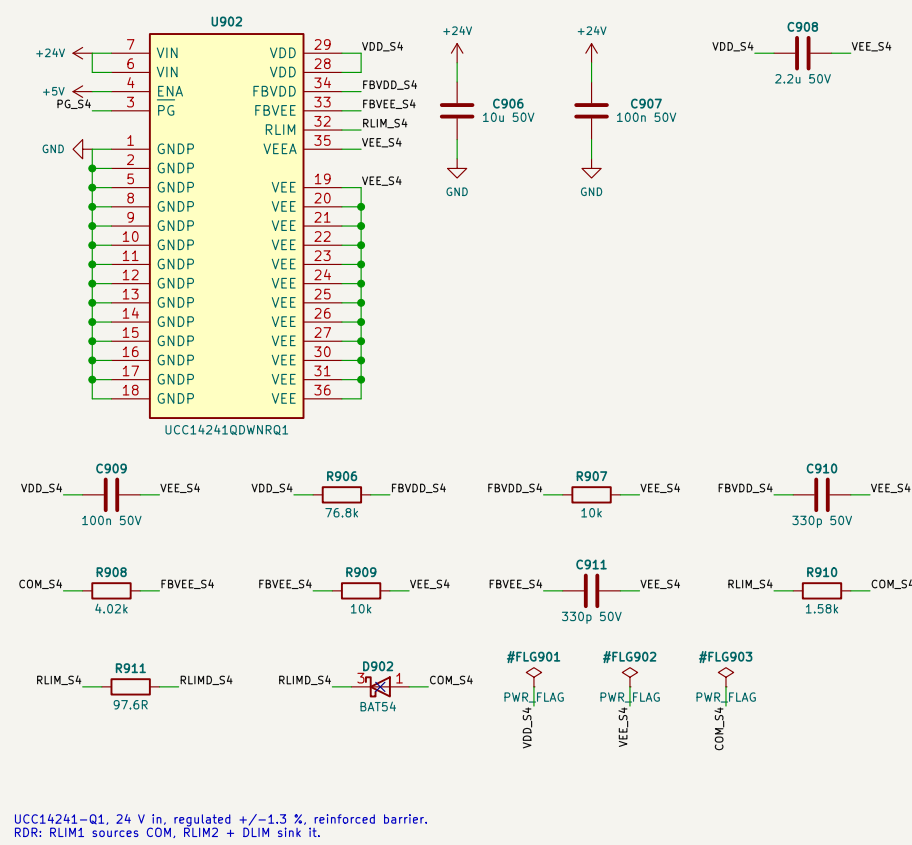


S4: LOGIC SIDE



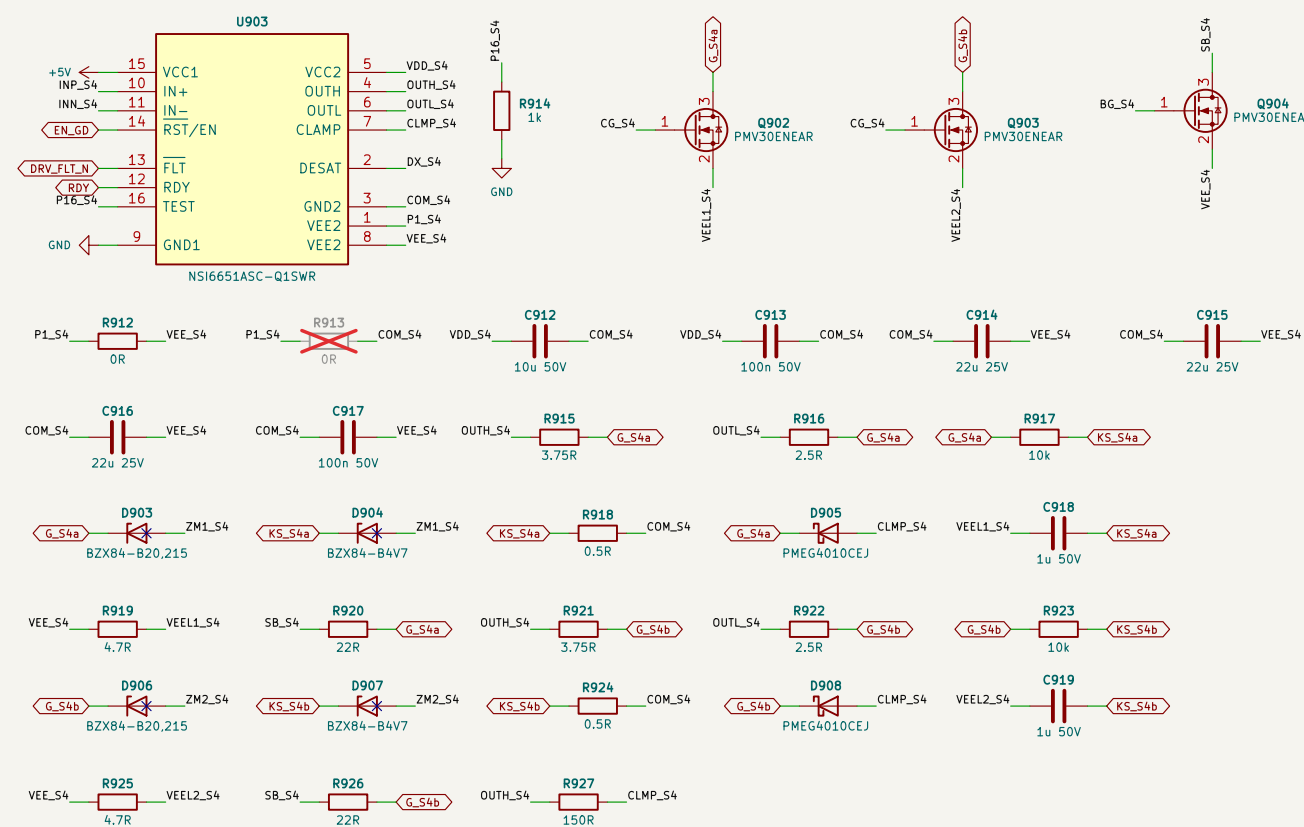
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

S4: ISOLATED BIAS +18 V / -3.5 V



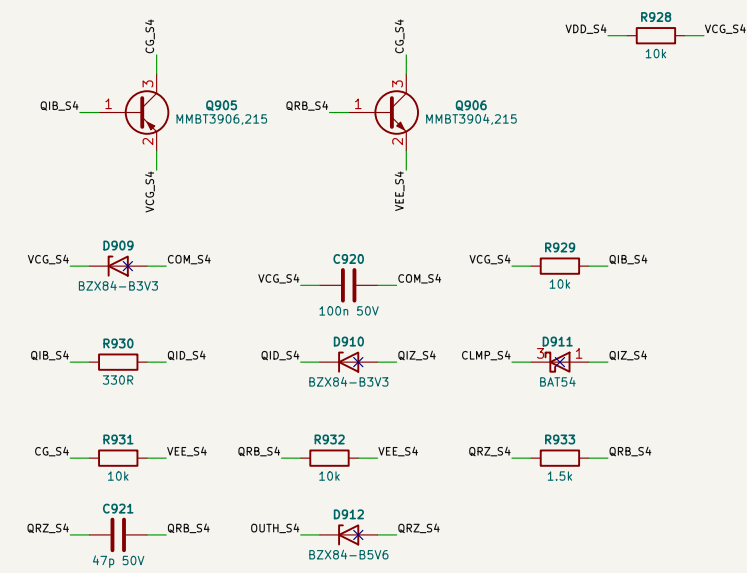
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

S4: DRIVER AND 2 GATES



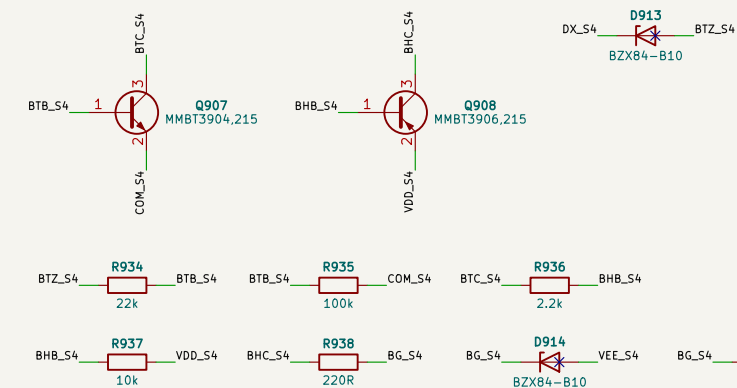
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

S4: MILLER CLAMP CONTROL



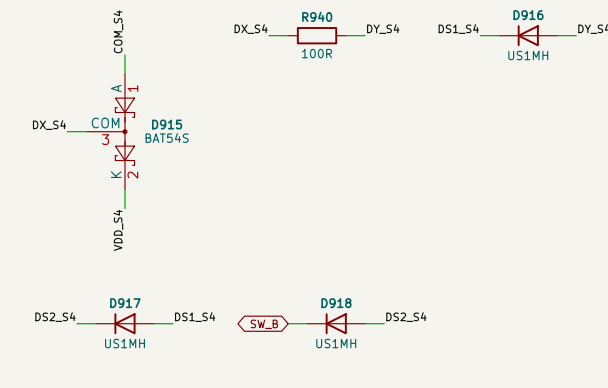
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V.
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

S4: LAYOUT RULES + MILLER MARGIN

SYNTEK: Kevin and Kelvin: 5Aa/5Ab as coupled pairs of equal length (+/-10 %) from the COM_54 star; Ron, Roff, Kelvin R. 10k; Zeners at each device; power sources joined only by the busbar. CLAMP LOOP: each AO3400A + C_VL at its device's gate-Kelvin pins, loop <= 1 nH, IC-22: milled spot between the drain pin and pins 2-4 of each TO-247-4L; DESAT string <= 2.5 nM, pad gap per diode: board coated to PD1. Miller hold (gdrv rev 5, ngpsic 1100 V / 72.4 A, 115 V/gs); die +1.41 V vs VGS(th) min 1.94 V at 175 C, margin 0.53 V (drn 4 PNP: +2.14 V) - BENCH TEST (double pulse, hot, minimum-threshold devices).